Sketch the vector function

$$\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2},$$

and compute its divergence.

Solution:

The divergence of the r-component in spherical coordinates is

$$\nabla \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r)$$

Using this, the divergence is $\nabla \cdot \boldsymbol{v} = 0$. From the sketch, clearly the divergence is not 0 at the origin. The problem arises at the point $\boldsymbol{r} = 0$ where $\boldsymbol{v} \to \infty$. At all other points $\boldsymbol{r} \neq 0$, the divergence is 0.

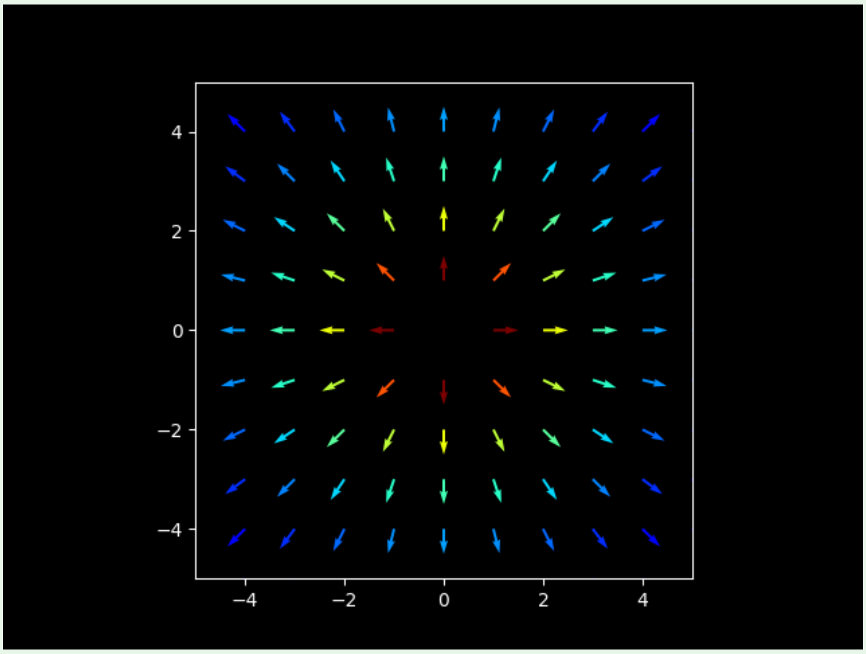

Figure 1: Sketch of the function $\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2}$

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